BADJI MOKHTAR UNIVERSITY ANNABA

FACULTY OF MEDICINE

CELLULAR PHYSIOLOGY

THE PLASMA MEMBRANE AND MEMBRANE TRANSPORT

Detailed Student Handout

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Learning objectives

By the end of this lecture, you should be able to:

1. Describe the structure and composition of the plasma membrane.
2. Explain the fluid mosaic model and the dynamic properties of the membrane.
3. State the roles of membrane lipids, proteins, cholesterol and the glycocalyx.
4. Identify and compare the mechanisms of membrane transport: passive, active and vesicular.
5. Explain the physical and energetic principles of transmembrane transport (gradients, ATP, coupling).
6. Relate membrane structure and transport to cell function and to clinical situations.

> **The thread of this course**
>
> **Structure → Properties → Transport → Function → Physiological and clinical relevance**
>
> In the diagrams: ↑ = movement **out** of the cell, ↓ = movement **into** the cell. Na^+ in red, K^+ in purple. Double lines = a membrane.

1. Definition and functions of the plasma membrane

The **plasma membrane** (cell membrane) is a thin biological membrane, about **7–10 nm** thick, that surrounds every living cell. It separates the **intracellular** environment (cytoplasm) from the **extracellular** environment. It is not a passive wall: it is a dynamic, highly organized structure and a **selective barrier** that controls exchanges between the cell and its surroundings.

Diagram 1. The four functions of the plasma membrane

1. Protection
Defines cell shape; barrier against mechanical stress, osmotic shock, toxins and pathogens.

2. Selective transport
Controls the movement of ions, nutrients and wastes, maintaining homeostasis.

3. Signalling and communication
Receives and transmits signals from chemical messengers such as hormones and neurotransmitters.

4. Cell recognition
Glycoproteins and glycolipids act as a cellular "ID"; they help the immune system tell self from non-self.

Clinical application: membrane integrity as a diagnostic principle

Cell injury → **Membrane disruption** loss of integrity → **Intracellular contents leak into the blood** → **Measurable biomarker**

Marker	Main clinical application
Troponin I / T	Myocardial injury; central biomarker of acute myocardial infarction
ALT / AST	Hepatocellular injury
CK / myoglobin	Skeletal muscle injury, rhabdomyolysis (risk of dangerous **hyperkalaemia** from K^+ released by muscle cells)
LDH	Non-specific: cell injury, tissue turnover, haemolysis

2. Molecular architecture of the membrane

The membrane is built from three components: **lipids** (phospholipids and cholesterol), **proteins** (integral and peripheral) and **carbohydrates** (the glycocalyx, on the outer surface only).

Diagram 2. Cross-section of the plasma membrane (fluid mosaic model)

Extracellular fluid	sugar chains (glycocalyx)	sugar chain			ions ↓	
Lipid bilayer	**Glycoprotein (integral)**	**Glycolipid**	Phospholipids: heads outside, tails inside	**Cholesterol between the tails**	**Channel protein (transmembrane)**	—
Cytoplasm						Peripheral protein (inner surface, linked to cytoskeleton)

Lipid rafts are small domains rich in cholesterol and sphingolipids that gather certain receptors and signalling proteins.

2.1 Phospholipids and the bilayer

Each phospholipid has a **hydrophilic head** (phosphate and an alcohol group: "water-loving") and **two hydrophobic tails** (fatty acid chains: "water-avoiding"). In water they form a **bilayer**: heads face the aqueous fluid on both sides and tails meet in the middle.

Diagram 3. Organization of the phospholipid bilayer

Extracellular fluid (water)	Aqueous environment
● ● ● **Heads (outer leaflet)**	**Hydrophilic**: face the water
‖ ‖ ‖ **Tails**	**Hydrophobic core**: barrier to ions and polar molecules
‖ ‖ ‖ **Tails**	
● ● ● **Heads (inner leaflet)**	**Hydrophilic**: face the cytoplasm
Cytoplasm (water)	Aqueous environment

Membrane asymmetry: the two leaflets differ

Phospholipid	Main leaflet	Particular role
Phosphatidylcholine (PC)	Outer	Major structural component
Sphingomyelin	Outer	Membrane stability; lipid rafts (with cholesterol)
Phosphatidylethanolamine (PE)	Inner	Membrane curvature; membrane fusion
Phosphatidylserine (PS)	Inner	Exposure on the outer surface = "eat-me" signal of **apoptosis**
Phosphatidylinositol (PI)	Inner	Precursor of second messengers (IP_3, DAG)

Where a phospholipid sits determines what it can do.

Movements of phospholipids

Lateral diffusion: fast
Moves sideways within the same leaflet; very frequent.

Rotation: fast
Spins around its own axis.

Flexion: fast
The tails bend while the head stays in place.

Flip-flop: very rare
Passes from one leaflet to the other. The polar head would have to cross the hydrophobic core; usually requires enzymes (flippases, scramblases).

2.2 Cholesterol

Cholesterol molecules are inserted **between phospholipids**: the small polar (OH) group lies near the heads and the rigid steroid ring and tail lie among the fatty acid tails.

Function	Explanation
Regulates fluidity (buffer)	↑ temperature: limits excessive fluidity. ↓ temperature: prevents the membrane from solidifying.
Decreases permeability	Reduces passage of ions and small polar molecules: a better barrier.
Modulates proteins	Influences the activity of membrane proteins and forms lipid rafts.

Clinical application: cholesterol

Atherosclerosis: LDL-derived cholesterol accumulates in arterial walls → atherosclerotic plaque.

Statins inhibit HMG-CoA reductase → ↓ hepatic cholesterol synthesis → ↑ hepatic LDL receptors → ↓ circulating LDL.

Key concept: cholesterol is essential for membrane function, but its metabolism must be tightly regulated.

2.3 Membrane proteins

- **Integral proteins** are embedded in the bilayer; many cross it completely (transmembrane proteins).
- **Peripheral proteins** sit on the inner or outer surface and do not cross the bilayer.

Function	Role
Channels (open or gated)	Let specific ions or water cross
Carriers (transporters) and pumps	Bind and move specific molecules
Receptors	Receive signals: hormones, neurotransmitters, drugs
Enzymes	Catalyse reactions at the membrane surface
Cell identity markers	Recognition of self versus non-self
Cell adhesion molecules (CAMs)	Attach cells to each other to form tissues
Structural proteins	Maintain shape; anchor the membrane to the cytoskeleton

2.4 Carbohydrates: the glycocalyx

Sugar chains attached to proteins (**glycoproteins**) or lipids (**glycolipids**) are found **only on the outer surface**. Together they form the **glycocalyx**, which serves for **recognition** (cellular ID card; blood groups are an example), **protection** and **adhesion**.

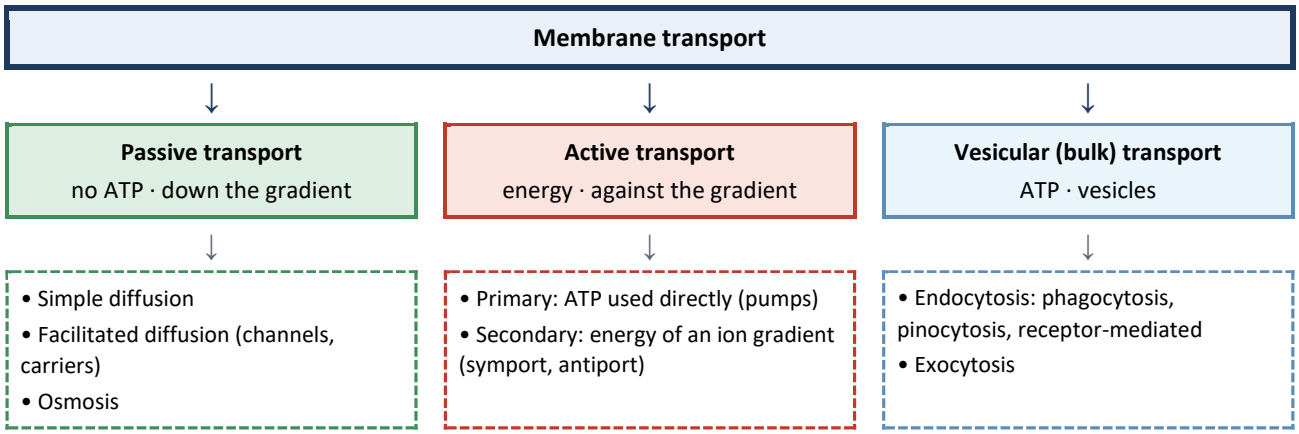
2.5 The fluid mosaic model

The membrane is a **mosaic** of proteins floating in a **fluid** lipid bilayer. Its components move constantly. This gives flexibility and self-repair, allows proteins to work, permits membrane fusion (cell division, vesicles) and keeps the right fluidity for transport and signalling. The membrane is also **asymmetric**: the outer face carries the glycocalyx and specific lipids and proteins; the inner face has different lipids and proteins and is linked to the cytoskeleton.

3. Membrane transport: overview

Membrane transport is the movement of substances (ions, nutrients, wastes, water) across the plasma membrane, into or out of the cell. The route depends on the size, charge and lipid solubility of the substance and on the membrane proteins available.

Diagram 4. Classification of membrane transport



Type	Energy	Direction	Examples
Passive	None (no ATP)	Down the gradient (high → low)	Simple and facilitated diffusion, osmosis
Active	Yes: ATP directly or an ion gradient	Against the gradient (low → high)	Na^+/K^+-ATPase, SGLT1 cotransport
Vesicular	Yes (ATP)	In (endocytosis) or out (exocytosis)	Phagocytosis, insulin secretion

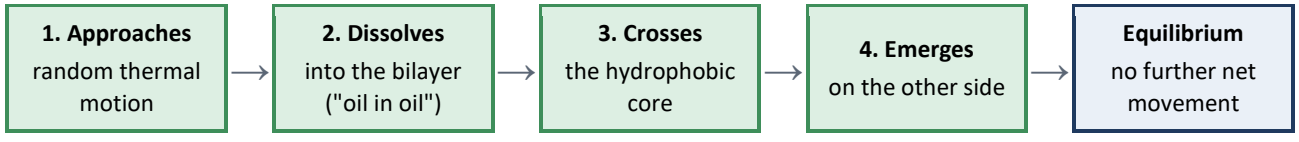
4. Passive transport

Passive transport needs **no ATP**: substances move **down** their concentration gradient (for ions, their electrochemical gradient), from high to low, using the kinetic energy of molecules. Net movement stops at equilibrium.

4.1 Simple diffusion

Molecules cross **directly through the phospholipid bilayer**, without any transport protein.

Diagram 5. Steps of simple diffusion



Molecules that use simple diffusion	Examples
Small, non-polar, lipid-soluble	Respiratory gases O_2 and CO_2; N_2
Steroid hormones	Cortisol, oestrogens, testosterone
Fat-soluble vitamins	A, D, E, K

Molecules that use simple diffusion	Examples
Small uncharged molecules	Ethanol, urea (slowly); water (slowly, mostly via aquaporins)
Lipid-soluble drugs	General anaesthetics; non-ionized forms of weak acids and bases (e.g. aspirin)

Rate of diffusion: Fick's law

Diffusion rate ∝ (surface area × concentration difference × permeability) / membrane thickness

Permeability increases with lipid solubility and decreases with molecular size; temperature increases molecular movement.

Factor	Effect on diffusion rate
Concentration gradient ↑	Faster
Temperature ↑	Faster
Surface area ↑	Faster (e.g. alveoli, intestinal microvilli)
Small, non-polar, lipid-soluble molecule	Faster
Membrane thickness ↑	Slower (e.g. pulmonary fibrosis, oedema)

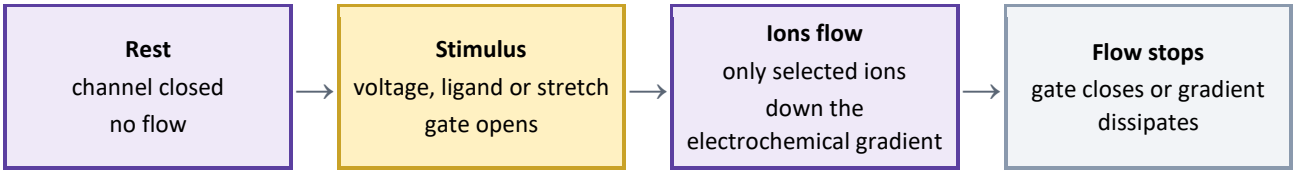
4.2 Facilitated diffusion

Many essential substances **cannot** cross the lipid core: they are too large (glucose, amino acids) or charged (Na^+, K^+, Ca^{2+}, Cl^-). They cross **down their gradient** through **specific transport proteins**, without ATP. There are two types: **channels** and **carriers**.

a) Channel-mediated diffusion

Channels form **water-filled pores**. They are highly **selective** (each type passes specific ions), often **gated**, and very **fast** (up to millions of ions per second).

Diagram 6. How a gated ion channel works

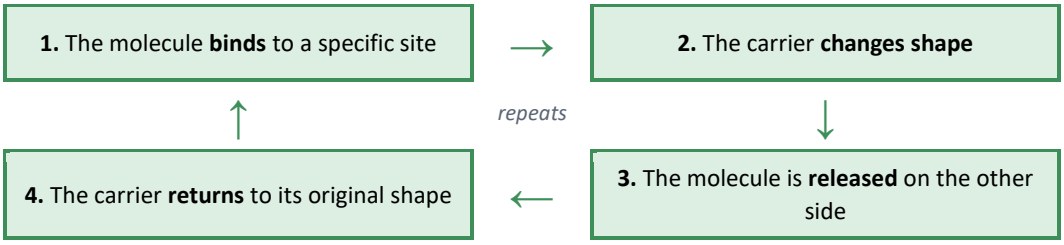


Type of gate	Opens in response to	Example
Voltage-gated	Change in membrane potential	Na^+ channels of the nerve impulse
Ligand-gated	Binding of a specific molecule	Nicotinic ACh receptor at the neuromuscular junction
Mechanically gated	Stretch or pressure	Touch receptors, hair cells of the inner ear

b) Carrier-mediated diffusion

Carriers (transporters) do not form a pore: they **bind** the molecule and **change shape** to release it on the other side. They are highly **specific**, slower than channels, can be **regulated**, and can be **saturated**. The main example is the **GLUT** family of glucose transporters.

Diagram 7. The carrier cycle



Feature	Simple diffusion	Channel	Carrier
Protein needed	No	Yes (pore)	Yes (binding site)
Specificity	Low	Selective for ions	Very high
Speed	Depends on lipid solubility	Very fast	Slower
Saturation	No: rate rises with the gradient	Rarely limiting	**Yes**: maximal rate (V_{max}) when all carriers are busy
Competition and regulation	No	Gating	Yes (e.g. insulin increases GLUT4 in muscle and fat)

4.3 Osmosis

Osmosis is the passive movement of **water** across a selectively permeable membrane, from the side with **low solute concentration** (more water) to the side with **high solute concentration** (less water). Water crosses slowly through the bilayer and rapidly through **aquaporins**, selective water channels that block ions and solutes.

Diagram 8. Direction of water movement in osmosis

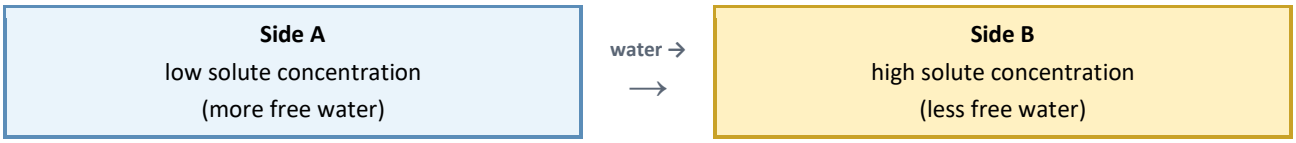
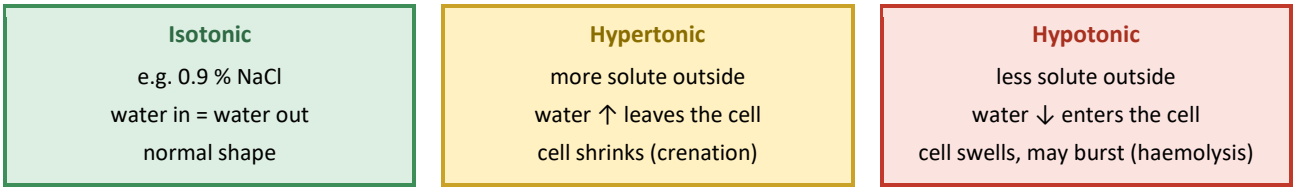


Diagram 9. Red blood cells in solutions of different tonicity



In the body: water balance between red blood cells and plasma, water reabsorption in the kidney (aquaporin-2, regulated by ADH), water absorption in the intestine. This is why intravenous fluids must have the right tonicity.

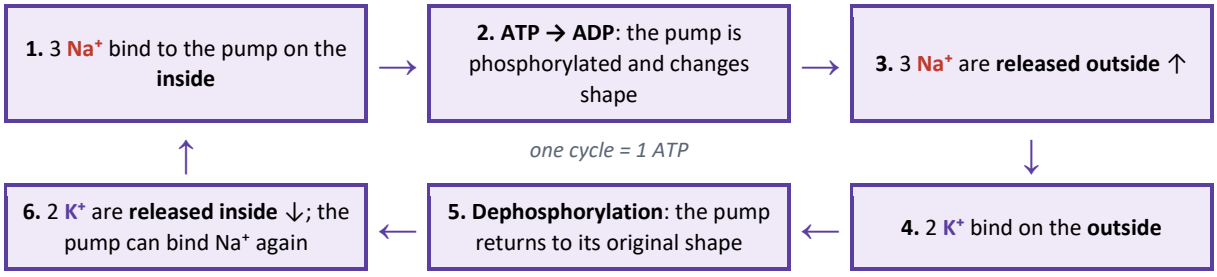
5. Active transport

Active transport moves substances **against** their concentration or electrochemical gradient (low → high). It needs **energy** and specific **transport proteins**. It maintains ion balance (Na^+, K^+, Ca^{2+}), pH and the membrane potential, allows nutrient absorption (glucose, amino acids, Ca^{2+}, Fe^{2+}), exports toxins and drugs (ABC transporters), and supports specialized functions such as gastric acid secretion.

5.1 Primary active transport

The transporter uses **direct hydrolysis of ATP** (it is an ATPase or "pump").

Diagram 10. The Na^+/K^+-ATPase cycle (3 Na^+ out, 2 K^+ in per ATP)



Read clockwise from box 1. Each cycle moves one net positive charge out: the pump is electrogenic.

Roles of the Na^+/K^+ pump: maintains the Na^+ and K^+ gradients and the resting potential, controls **cell volume**, and creates the Na^+ gradient used by **secondary active transport**.

Pump	Moves	Location	Role and clinical link
Na^+/K^+-ATPase	3 Na^+ out, 2 K^+ in	All cells (basolateral side in epithelia)	Gradients, volume, resting potential. Inhibited by **digoxin**
Ca^{2+}-ATPase	Ca^{2+} out of the cytosol	Plasma membrane (PMCA); sarcoplasmic reticulum (SERCA)	Keeps cytosolic Ca^{2+} very low; SERCA pumps Ca^{2+} back into the SR so muscle can **relax**
H^+/K^+-ATPase ("acid producer")	H^+ out, K^+ in (1:1)	Gastric parietal cells; distal nephron	$H^+ + Cl^- \rightarrow HCl$. Stimulated by histamine, ACh, gastrin; inhibited by somatostatin. Blocked by **proton pump inhibitors** (omeprazole): ulcers, GERD, Zollinger–Ellison
ABC transporters	Drugs, toxins, lipids, Cl^-	Many cells	Multidrug resistance (P-glycoprotein); CFTR, a Cl^- channel of this family, is defective in **cystic fibrosis**

5.2 Secondary active transport (cotransport)

The transporter uses the **energy stored in an ion gradient**, usually the Na^+ gradient created by the Na^+/K^+ pump. The ion moving **down** its gradient drives another substance **against** its gradient (coupling). The transporter itself does not use ATP, but stops working if the pump stops.

Diagram 11. Symport and antiport

Symport (cotransport)	Antiport (exchange)
ion and substance move in the **same direction**	ion and substance move in **opposite directions**
Na^+ ↓ + glucose ↓	Na^+ ↓ / H^+ ↑
SGLT1, SGLT2; Na^+-amino acids; Na^+-K^+-$2Cl^-$ (NKCC); Na^+-bile salts; Na^+-neurotransmitter reuptake	Na^+/H^+ exchanger (NHE: pH); Na^+/Ca^{2+} exchanger (NCX: removes Ca^{2+})

Diagram 12. Glucose absorption across an intestinal (or renal tubular) cell

Lumen	glucose (low)	Na^+ (high)
Apical membrane	**SGLT1** (symport): 2 Na^+ + 1 glucose ↓ · secondary active	
Cell cytoplasm	glucose (high) ↓	Na^+ kept low ↓
Basolateral membrane	**GLUT2**: facilitated diffusion ↓	**Na^+/K^+-ATPase**: 3 Na^+ out, 2 K^+ in (ATP)
Interstitium and blood	glucose → blood	Na^+ → blood

One cell uses all three mechanisms: secondary active (SGLT1), passive (GLUT2) and primary active (pump) transport.

Clinical links

Oral rehydration solution contains glucose and Na^+ together: SGLT1 absorbs both, and water follows by osmosis. This is why it works in cholera and infant diarrhoea.

SGLT2 inhibitors (gliflozins) block glucose reabsorption in the proximal tubule: they are used in type 2 diabetes and heart failure.

6. Vesicular (bulk) transport

Bulk transport moves **large molecules, particles or volumes of fluid** inside membrane **vesicles**. It is needed when molecules are too big for transporters (proteins, polysaccharides), to take up particles (bacteria, debris) and for bulk secretion (hormones, enzymes, neurotransmitters). It **requires ATP**, is **highly regulated** and works in both directions.

6.1 Endocytosis

Invagination the membrane folds inward around the material → **Vesicle formation** the membrane pinches off → **Trafficking** the vesicle moves into the cell for processing

Diagram 13. Phagocytosis ("cell eating")

Pseudopods surround the particle → **Phagosome** vesicle containing the particle → **Fusion with a lysosome** = phagolysosome → **Degradation** enzymes and reactive oxygen species

Professional phagocytes: neutrophils, macrophages, dendritic cells. Roles: clearance of pathogens, dead cells and foreign particles.

Diagram 14. Receptor-mediated endocytosis ("selective uptake")

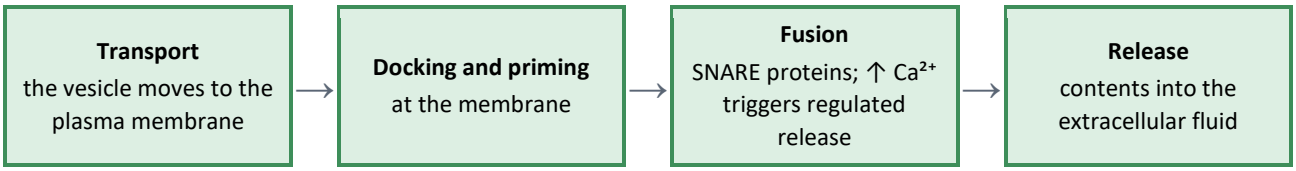
1. Binding ligand binds its receptor → **2. Coated pit** receptors cluster; clathrin coats the pit → **3. Coated vesicle** dynamin pinches it off → **4. Uncoating** delivery to the early endosome → **5. Sorting** ligand degraded or used; receptor recycled to the surface

	Phagocytosis	Pinocytosis	Receptor-mediated
Material	Large solid particles (bacteria, debris)	Extracellular fluid and dissolved solutes	Specific ligands (LDL, transferrin-iron, some vitamins)
Selectivity	Selective (recognition)	**Non-selective**	**Highly selective**, saturable
Vesicle	Large phagosome	Small (≈ 0.1 µm)	Clathrin-coated vesicle
Cells	Neutrophils, macrophages, dendritic cells	Almost all cells, continuously	Many cells (e.g. hepatocytes for LDL)

> **Clinical link: familial hypercholesterolaemia**
>
> Mutations of the **LDL receptor** prevent receptor-mediated uptake of LDL: cholesterol stays in the blood and causes early atherosclerosis.

6.2 Exocytosis

Diagram 15. Steps of exocytosis



Examples: neurotransmitter release at synapses, insulin secretion by pancreatic β cells, digestive enzymes from acinar cells. Exocytosis also adds membrane to the cell surface.

7. Summary

Mechanism	Energy	Direction	Protein or vesicle	Example
Simple diffusion	None	Down the gradient	None	O_2, CO_2, steroids
Channel	None	Down the electrochemical gradient	Channel (often gated)	Na^+, K^+ channels; aquaporins
Carrier (facilitated)	None	Down the gradient	Carrier (saturable)	GLUT
Osmosis	None	Towards higher solute concentration	Bilayer or aquaporins	Water
Primary active	ATP directly	Against the gradient	Pump (ATPase)	Na^+/K^+-ATPase, H^+/K^+-ATPase
Secondary active	Ion gradient	Against the gradient (driven substance)	Symporter or antiporter	SGLT1, NHE, NCX
Endocytosis	ATP	Into the cell	Vesicle	Phagocytosis, LDL uptake
Exocytosis	ATP	Out of the cell	Vesicle	Insulin, neurotransmitters

Key takeaways

- The plasma membrane is a selective, dynamic interface: a fluid, asymmetric bilayer carrying proteins and a glycocalyx.
- Lipids form the barrier; proteins provide transport, signalling, adhesion and identity; cholesterol buffers fluidity.
- **Passive** transport follows gradients without ATP; **active** transport works against gradients using ATP directly (primary) or an ion gradient (secondary); **vesicular** transport moves large cargo in vesicles.
- The Na^+/K^+ pump underlies the resting potential, cell volume and all Na^+-coupled transport.
- Transport defects cause disease, and transporters are major drug targets (PPIs, digoxin, gliflozins, statins via LDL receptors).

Abbreviations

Abbr.	Meaning	Abbr.	Meaning
GLUT	Glucose transporter (facilitated)	**SGLT**	Sodium-glucose linked transporter
NHE	Na^+/H^+ exchanger	**NCX**	Na^+/Ca^{2+} exchanger
NKCC	Na^+-K^+-$2Cl^-$ cotransporter	**ABC**	ATP-binding cassette transporter
PMCA / SERCA	Plasma membrane / sarco-endoplasmic reticulum Ca^{2+}-ATPase	**PPI**	Proton pump inhibitor
LDL	Low-density lipoprotein	**SNARE**	Proteins that mediate vesicle fusion

The Plasma Membrane and Membrane Transport

English–Arabic terminology | First-year medical physiology

Arabic equivalents are provided for the meaning used in this course.

Key terms

English	العربية
Plasma membrane	الغشاء البلازمي
Cell membrane	غشاء الخلية
Intracellular fluid	السائل داخل الخلية
Extracellular fluid	السائل خارج الخلية
Selective permeability	النفاذية الانتقائية
Homeostasis	الاستتباب
Phospholipid	الفوسفوليبيد
Lipid bilayer	الطبقة الدهنية الثنائية
Hydrophilic head	الرأس المحب للماء
Hydrophobic tail	الذيل الكاره للماء
Membrane leaflet	وريقة الغشاء
Membrane asymmetry	عدم تماثل الغشاء
Phosphatidylcholine	فوسفاتيديل كولين
Sphingomyelin	السفينغومَيلين
Phosphatidylethanolamine	فوسفاتيديل إيثانول أمين
Phosphatidylserine	فوسفاتيديل سيرين
Phosphatidylinositol	فوسفاتيديل إينوزيتول
Cholesterol	الكوليسترول
Membrane fluidity	سيولة الغشاء
Lateral diffusion	الانتشار الجانبي
Flip-flop	الانتقال بين وريقتي الغشاء
Integral protein	بروتين غشائي متكامل
Peripheral protein	بروتين غشائي محيطي
Glycoprotein	بروتين سكري
Glycolipid	دهن سكري
Glycocalyx	الغلاف السكري
Fluid mosaic model	نموذج الفسيفساء المائع
Receptor	مستقبل
Cell adhesion	التصاق الخلايا
Concentration gradient	تدرّج التركيز
Electrochemical gradient	التدرّج الكهروكيميائي
Passive transport	النقل السلبي
Simple diffusion	الانتشار البسيط
Facilitated diffusion	الانتشار المُيَسَّر
Ion channel	قناة أيونية
Gated channel	قناة ذات بوابة
Carrier protein	بروتين ناقل
Osmosis	التناضح

Aquaporin	قناة مائية (أكوابورين)
Tonicity	التوترية
Hypotonic solution	محلول ناقص التوتر
Hypertonic solution	محلول مفرط التوتر
Active transport	النقل النشط
Primary active transport	النقل النشط الأولي
Secondary active transport	النقل النشط الثانوي
Sodium–potassium pump	مضخة الصوديوم والبوتاسيوم
Calcium pump	مضخة الكالسيوم
Proton pump	مضخة البروتونات
Symport	النقل المشترك في الاتجاه نفسه
Antiport	النقل التبادلي في اتجاهين متعاكسين
Sodium–glucose cotransporter	الناقل المشترك للصوديوم والغلوكوز
Vesicular transport	النقل الحويصلي
Endocytosis	الإدخال الخلوي
Exocytosis	الإخراج الخلوي
Phagocytosis	البلعمة
Pinocytosis	الاحتساء الخلوي
Receptor-mediated endocytosis	الإدخال الخلوي بوساطة المستقبلات
Vesicle	حويصلة
Phagosome	جسيم بلعمي
Lysosome	الجسيم الحال
Clathrin-coated pit	حفرة مكسوّة بالكلاثرين
Membrane fusion	اندماج الأغشية
Atherosclerosis	تصلّب الشرايين

Useful verbs

English	العربية
To surround	يحيط بـ
To separate	يفصل
To protect	يحمي
To regulate	ينظّم
To bind	يرتبط بـ
To cross	يعبر
To diffuse	ينتشر
To dissolve	يذوب
To transport	ينقل
To pump	يضخّ
To absorb	يمتصّ
To secrete	يفرز
To engulf	يبتلع أو يحيط بـ
To fuse	يندمج
To release	يحرّر أو يطلق
To maintain	يحافظ على